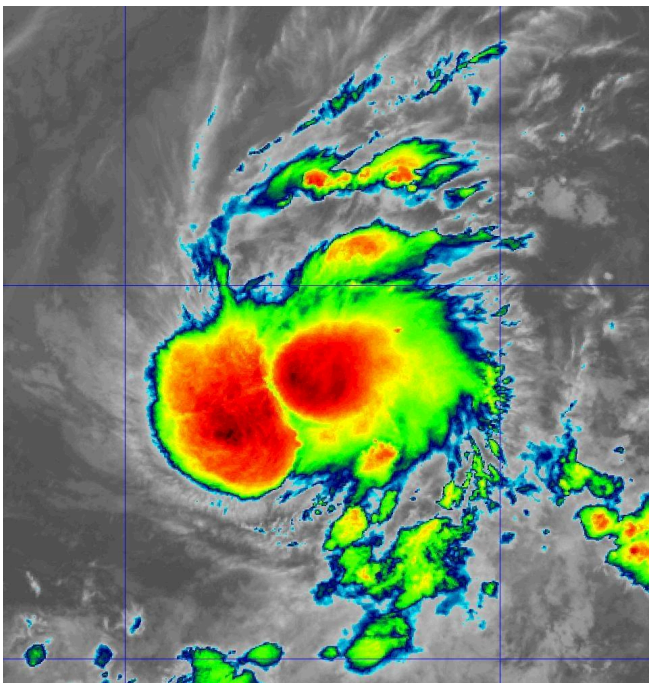
NATIONAL HURRICANE CENTER TROPICAL CYCLONE REPORT[1]

TROPICAL STORM SONIA

(EP182025)

24–28 October 2025

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National Hurricane Center
5 December 2025



GOES-WEST INFRARED SATELLITE IMAGE OF TROPICAL STORM SONIA AT 1200 UTC 25 OCTOBER 2025. IMAGE COURTESY NOAA/NESDIS/STAR.

Sonia was a tropical storm that moved over central and western portions of the eastern Pacific basin and did not impact land.

[1] This is an abbreviated Tropical Cyclone Report since there were no coastal watches or warnings issued and no direct fatalities reported in association with Sonia.

Tropical Storm Sonia

24–28 October 2025

BEST TRACK

The "best track[2]" positions and intensities for Tropical Storm Sonia are listed in Table 1. The "best track" chart of Sonia's path is given in Fig. 1, with the wind and pressure histories along with available observations[3] shown in Figs. 2 and 3, respectively.

There were no ship reports of tropical-storm-force winds associated with Sonia.

Origin

Sonia originated from a tropical wave that moved across Central America on 16–17 October. The wave entered the eastern Pacific basin on 18 October and moved westward at about 10 kt over the next several days while producing disorganized showers and thunderstorms. A surface trough became evident in scatterometer data on 22–23 October, and convection gradually organized along this feature while the low-level circulation became tighter and better defined. It is estimated that a tropical depression formed by 0600 UTC 24 October, about 615 n mi south-southwest of the southern tip of the Baja California peninsula.

Peak Intensity and Minimum Pressure

Sonia's peak intensity of 45 kt from 1200 UTC 25 October to 1200 UTC 26 October is supported by T3.0/45-kt Dvorak classifications from TAFB and a blend of UW-CIMSS SATCON, AiDT, and D-MINT estimates that peaked between 44–48 kt during this period. A 0548 UTC 26 October ASCAT-C pass showed peak winds of 41 kt, which also supports a 45-kt intensity estimate after accounting for undersampling due to the coarse resolution of the instrument.

The estimated minimum central pressure of 1001 mb is based on pressure estimates from TAFB Dvorak classifications and the Knaff-Zehr-Courtney pressure-wind relationship.

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[2] A digital record of the complete best track, including wind radii, can be found on line at ftp://ftp.nhc.noaa.gov/atcf. Data for the current year's storms are located in the *btk* directory, while previous years' data are located in the *archive* directory.

[3] Observations include subjective satellite-based Dvorak technique intensity estimates from the Tropical Analysis and Forecast Branch (TAFB) and the Satellite Analysis Branch (SAB), objective Advanced Dvorak Technique (ADT) estimates and Satellite Consensus (SATCON) estimates from the Cooperative Institute for Meteorological Satellite Studies/University of Wisconsin-Madison. Data and imagery from NOAA polar-orbiting satellites including the Advanced Microwave Sounding Unit (AMSU), the NASA Global Precipitation Mission (GPM), the European Space Agency's Advanced Scatterometer (ASCAT), the Defense Meteorological Satellite Program (DMSP) satellites, and the Time-Resolved Observations of Precipitation structure and storm Intensity with a Constellation of Smallsats (TROPICS) satellites, among others, were also useful in constructing the best track of Sonia.

CASUALTY AND DAMAGE STATISTICS

There were no reports of damage or casualties associated with Sonia.

FORECAST AND WARNING VERIFICATION

Table 2 provides the number of hours in advance of formation for the first NHC Tropical Weather Outlook (TWO) forecast in each likelihood category. The 7-day NHC genesis forecasts provided sufficient lead times. The 2-day forecasts were not as skillful, as Sonia did not reach the high category before genesis occurred. Figure 4 shows that all of the 7-day TWO genesis areas captured the location where Sonia formed.

A verification of NHC official track forecasts for Sonia is given in Table 3a. Official track forecast errors were lower than the mean official errors for the previous 5-yr period at all forecast times. A homogeneous comparison of the official track errors with selected guidance models is given in Table 3b. The NHC forecasts performed very well, beating the individual models at most forecast times and remaining competitive with the consensus aids.

A verification of NHC official intensity forecasts for Sonia is given in Table 4a. Official intensity forecast errors were lower than or comparable to the mean official errors for the previous 5-yr period at all forecast times except 36 h. A homogeneous comparison of the official intensity errors with selected guidance models is given in Table 4b. The ECMWF (EMXI) and Google DeepMind (GDMI) were the top performing intensity models.

There were no coastal watches or warnings issued for Sonia.

ACKNOWLEDGEMENTS

Dr. Philippe Papin (NHC) created the GTWO verification graphic (Fig. 4).

Table 1. Best track for Tropical Storm Sonia, 24–28 October 2025.

Date/Time (UTC)	Latitude (°N)	Longitude (°W)	Pressure (mb)	Wind Speed (kt)	Stage
24 / 0600	13.1	113.4	1007	30	tropical depression
24 / 1200	13.2	114.5	1007	30	"
24 / 1800	13.2	115.6	1006	30	"
25 / 0000	13.2	116.4	1005	35	tropical storm
25 / 0600	13.3	117.2	1004	40	"
25 / 1200	13.4	117.9	1002	45	"
25 / 1800	13.4	118.7	1001	45	"
26 / 0000	13.4	119.0	1001	45	"
26 / 0600	13.4	119.2	1002	45	"
26 / 1200	13.5	119.4	1002	45	"
26 / 1800	13.7	119.5	1004	40	"
27 / 0000	13.9	119.7	1004	40	"
27 / 0600	14.0	120.3	1004	40	"
27 / 1200	13.9	120.8	1005	35	"
27 / 1800	14.0	121.3	1005	35	"
28 / 0000	14.3	121.8	1005	35	"
28 / 0600	14.7	122.4	1004	40	"
28 / 1200	15.0	123.1	1004	40	"
28 / 1800	15.2	123.9	1006	35	"
29 / 0000	15.3	124.8	1007	30	low
29 / 0600	15.3	125.6	1008	30	"
29 / 1200	15.4	126.6	1009	25	"
29 / 1800	15.5	127.8	1009	25	"
30 / 0000					dissipated
25 / 1800	13.4	118.7	1001	45	minimum pressure and maximum winds

Table 2. Number of hours in advance of formation associated with the first NHC Tropical Weather Outlook forecast in the indicated likelihood category. Note that the timings for the "Low" category do not include forecasts of a 0% chance of genesis.

	Hours Before Genesis	
	48-Hour Outlook	**168-Hour Outlook**
Low (<40%)	132	132
Medium (40%-60%)	24	114
High (>60%)	–	72

Table 3a. NHC official (OFCL) and climatology-persistence skill baseline (OCD5) track forecast errors (n mi) for Tropical Storm Sonia, 24–28 October 2025. Mean errors for the previous 5-yr period are shown for comparison. Official errors that are smaller than the 5-yr means are shown in boldface type.

	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	**20.1**	**27.2**	**27.8**	**33.7**	**36.1**	**33.5**	**13.3**	
OCD5	32.4	67.3	107.7	138.6	172.6	216.7	254.7	
Forecasts	15	13	11	9	7	5	1	
OFCL (2020-24)	22.4	33.6	43.5	53.2	65.1	79.4	108.6	135.2
OCD5 (2020-24)	38.2	75.3	117.2	161.0	206.0	252.0	340.7	422.7

Table 3b. Homogeneous comparison of selected track forecast guidance models (in n mi) for Tropical Storm Sonia, 24–28 October 2025. Errors smaller than the NHC official forecast are shown in boldface type. The number of official forecasts shown here will generally be smaller than that shown in Table 3a due to the homogeneity requirement.

Model ID	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	21.1	28.8	28.7	33.4	38.1	32.5		
OCD5	32.5	67.6	103.6	129.3	159.0	198.5		
GFSI	**20.4**	29.9	41.2	58.5	79.6	81.3		
HWFI	22.2	34.7	42.1	70.1	119.0	146.5		
HMNI	**20.9**	**28.7**	36.6	57.7	87.3	114.3		
HFAI	**19.5**	**22.3**	35.9	56.7	80.3	90.2		
HFBI	**18.8**	**27.6**	36.6	59.1	73.1	85.2		
EMXI	23.4	45.2	63.2	66.9	70.0	69.5		
CMCI	25.1	39.3	59.3	88.3	113.6	141.0		
CTCI	21.6	33.9	39.6	50.7	69.9	59.3		
GDMI	21.2	32.8	33.7	35.0	41.3	38.3		
TVCE	**17.8**	**22.3**	**26.6**	38.0	60.8	69.0		
TVCX	**18.2**	**22.8**	28.8	40.3	59.8	66.6		
GFEX	**17.9**	29.3	36.1	40.4	47.2	46.6		
TVDG	**17.7**	**23.4**	**27.3**	39.1	60.1	66.9		
HCCA	**17.9**	**27.2**	**28.4**	34.7	40.9	48.5		
FSSE	**17.6**	**27.9**	**27.9**	**28.1**	44.0	45.4		
AEMI	21.9	30.4	32.1	45.3	62.0	56.5		
TABS	25.0	35.3	36.8	40.8	45.8	66.9		
TABM	27.3	43.7	60.3	52.7	48.4	**25.8**		
TABD	38.7	77.4	120.6	147.5	173.7	190.4		
Forecasts	14	12	10	8	6	4		

Table 4a. NHC official (OFCL) and climatology-persistence skill baseline (OCD5) intensity forecast errors (kt) for Tropical Storm Sonia, 24–28 October 2025. Mean errors for the previous 5-yr period are shown for comparison. Official errors that are smaller than the 5-yr means are shown in boldface type.

	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	5.7	**7.7**	11.4	**11.7**	**10.0**	**8.0**	**5.0**	
OCD5	6.0	6.6	9.7	10.4	16.7	16.8	15.0	
Forecasts	15	13	11	9	7	5	1	
OFCL (2020-24)	5.7	8.9	10.8	12.9	14.4	15.5	17.0	18.6
OCD5 (2020-24)	7.4	12.4	16.2	19.2	21.0	21.3	21.5	21.6

Table 4b. Homogeneous comparison of selected intensity forecast guidance models (in kt) for Tropical Storm Sonia, 24–28 October 2025. Errors smaller than the NHC official forecast are shown in boldface type. The number of official forecasts shown here will generally be smaller than that shown in Table 4a due to the homogeneity requirement.

Model ID	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	5.7	7.5	12.0	12.5	10.8	6.2		
OCD5	5.9	**6.3**	**10.1**	**11.6**	18.3	18.0		
HWFI	**3.4**	**6.6**	**11.5**	17.1	15.5	8.5		
HMNI	**4.6**	**7.4**	**10.7**	13.4	12.3	**4.5**		
HFAI	**5.4**	7.8	**9.5**	**10.0**	**7.3**	**6.0**		
HFBI	**3.0**	**3.9**	**7.6**	**11.0**	15.0	10.8		
DSHP	**5.6**	9.1	12.9	12.9	**9.2**	6.2		
LGEM	5.8	8.5	**11.0**	**11.4**	**7.7**	**5.5**		
ICON	**4.6**	**7.3**	**11.6**	13.2	**10.5**	**5.2**		
IVCN	**4.5**	**6.6**	**10.6**	**11.8**	**9.8**	**5.2**		
IVDR	**4.5**	**6.5**	**10.2**	**11.6**	**10.3**	**5.2**		
CTCI	6.0	8.4	12.4	13.0	**8.3**	**3.2**		
GFSI	**5.6**	7.8	**8.6**	**8.1**	**8.7**	6.8		
EMXI	**4.4**	**5.1**	**5.3**	**3.4**	**3.3**	**2.5**		
GDMI	**4.2**	**3.9**	**5.0**	**4.6**	**4.8**	**4.2**		
HCCA	**4.5**	**7.2**	12.9	16.4	13.8	8.0		
FSSE	**4.9**	7.5	**11.5**	**12.4**	**10.7**	**4.5**		
Forecasts	14	12	10	8	6	4		

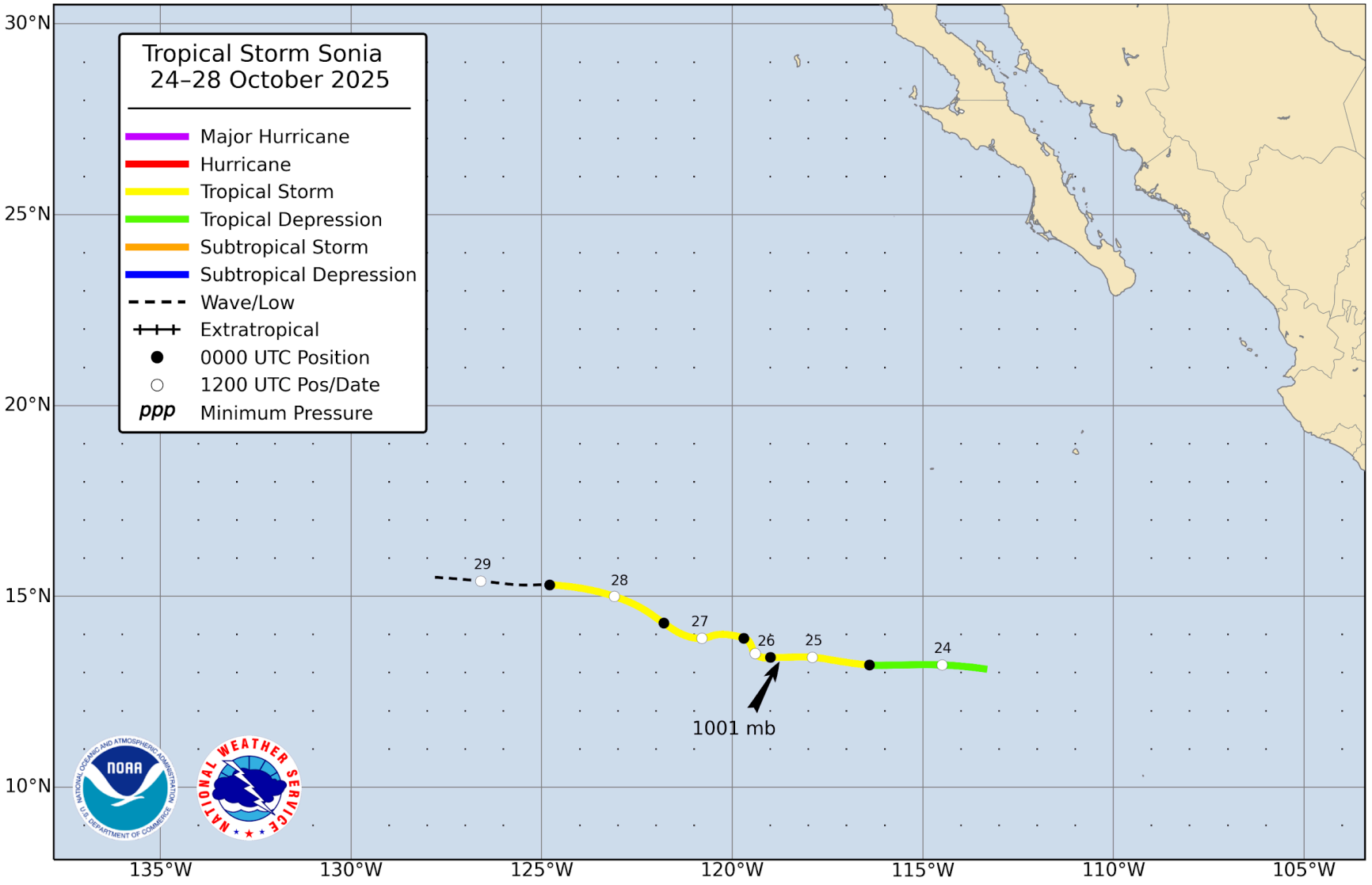


Figure 1. Best track positions for Tropical Storm Sonia, 24–28 October 2025.

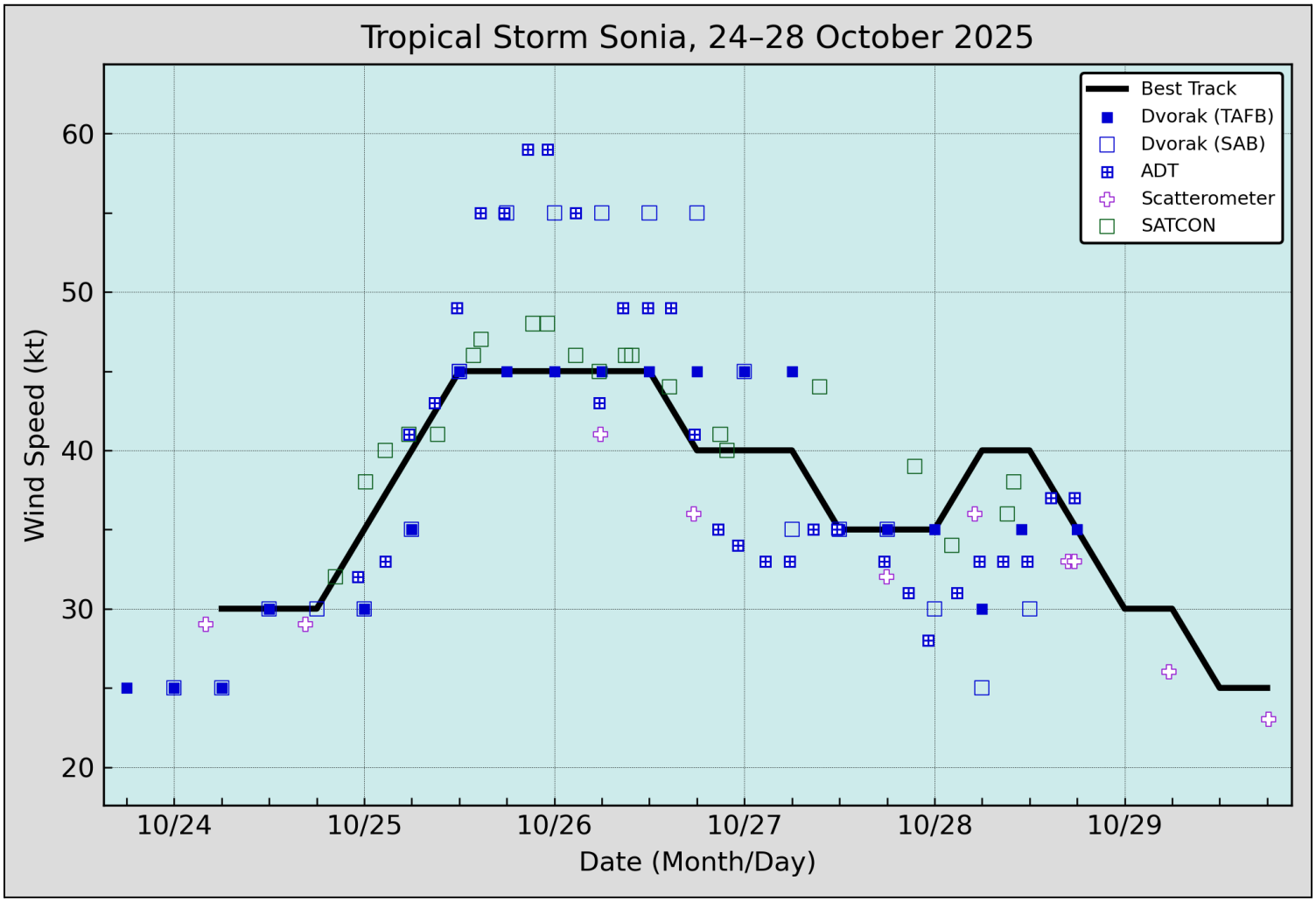


Figure 2. Selected wind observations and best track maximum sustained surface wind speed curve for Tropical Storm Sonia, 24–28 October 2025. Advanced Dvorak Technique estimates represent the Current Intensity at the nominal observation time. SATCON intensity estimates are from the Cooperative Institute for Meteorological Satellite Studies. Dashed vertical lines correspond to 0000 UTC.

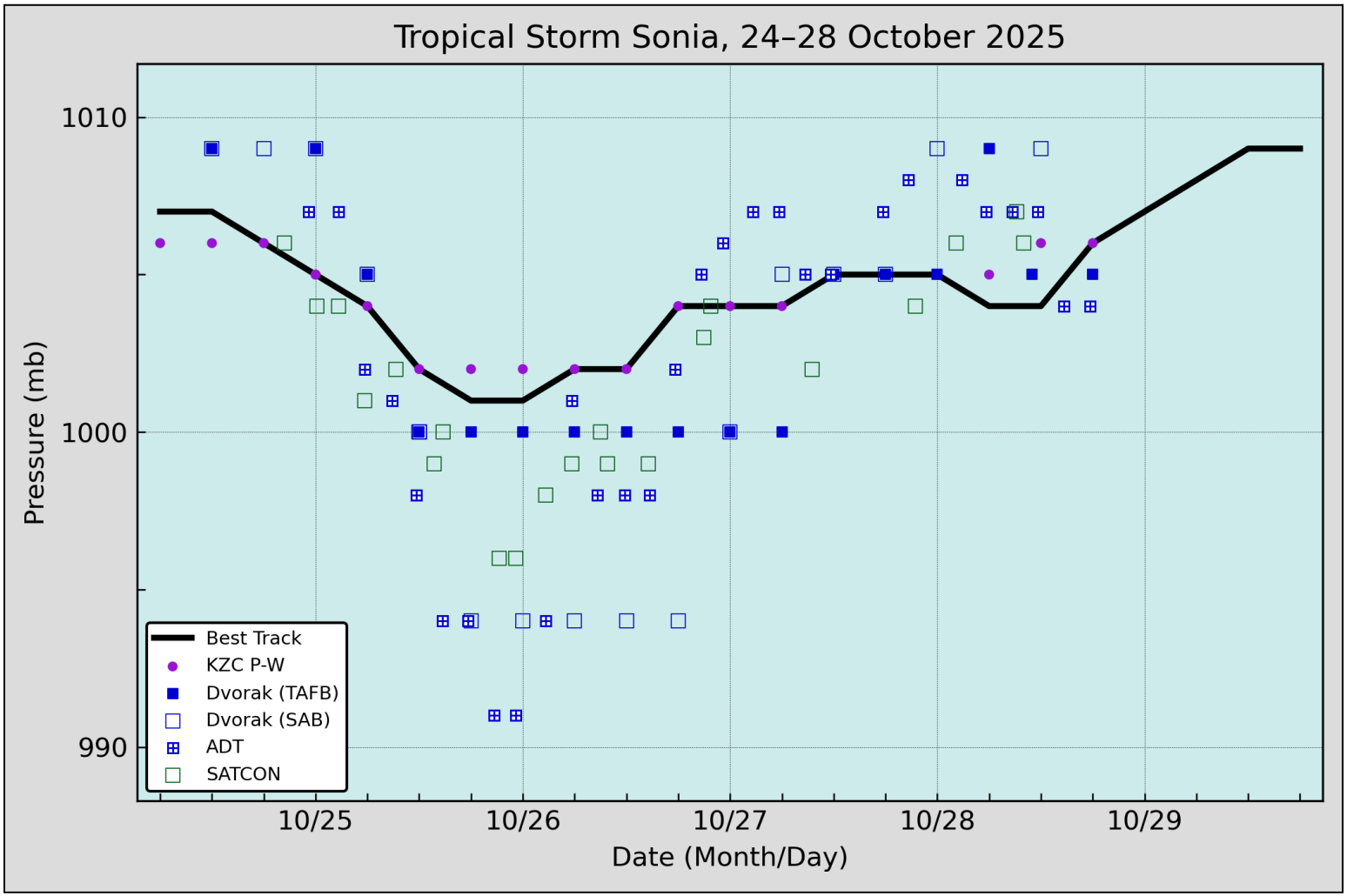


Figure 3. Selected pressure observations and best track minimum central pressure curve for Tropical Storm Sonia, 24–28 October 2025. Advanced Dvorak Technique estimates represent the Current Intensity at the nominal observation time. SATCON intensity estimates are from the Cooperative Institute for Meteorological Satellite Studies. KZC P-W refers to pressure estimates derived using the Knaff-Zehr-Courtney pressure-wind relationship. Dashed vertical lines correspond to 0000 UTC.

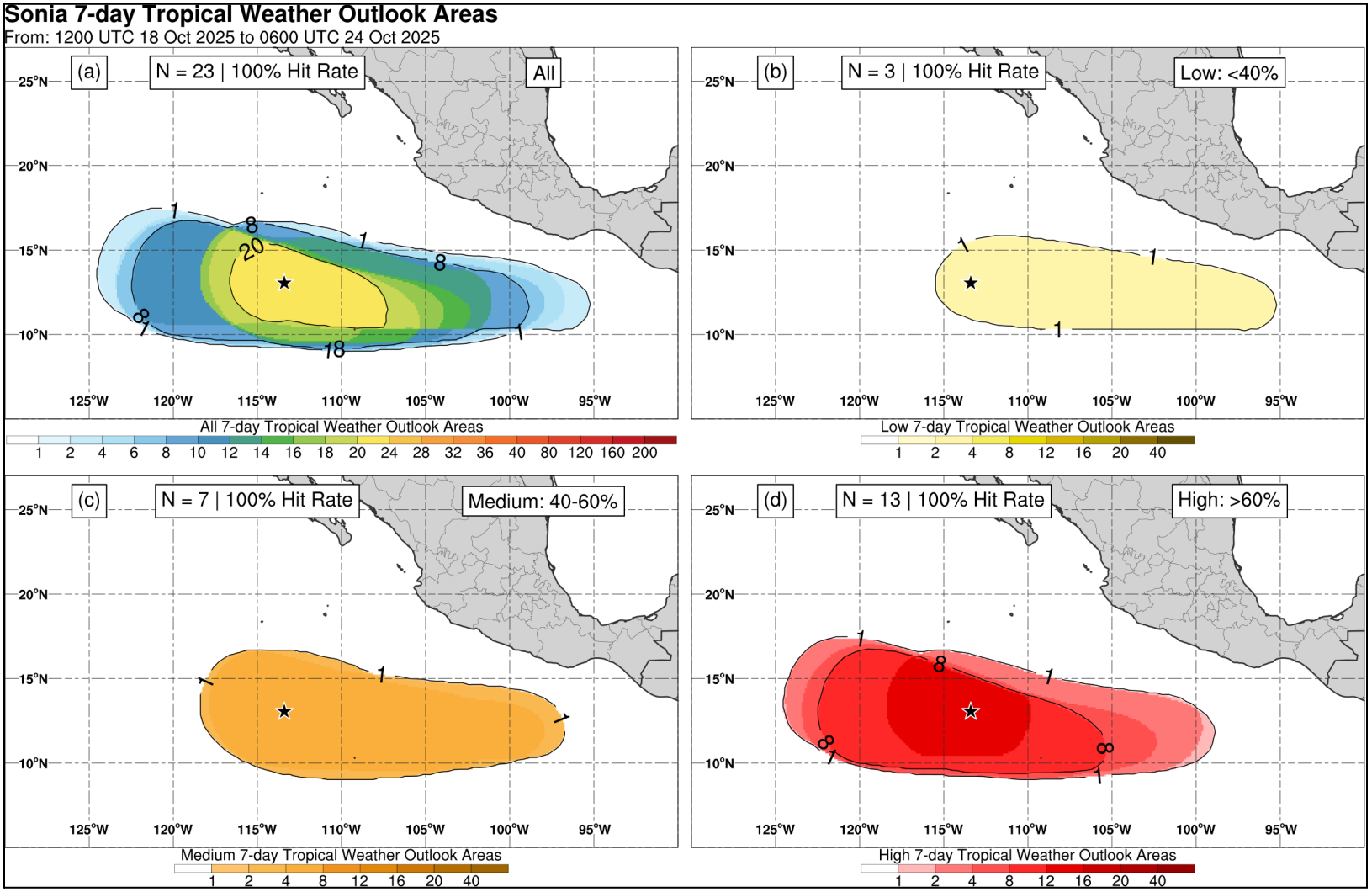


Figure 4. Composites of 7-day tropical cyclone genesis areas depicted in NHC's Tropical Weather Outlooks prior to the formation of Sonia for (a) all probabilistic genesis categories, (b) the low (<40%) category, (c) medium (40–60%) category, and (d) high (>60%) category. The location of genesis is indicated by the black star.